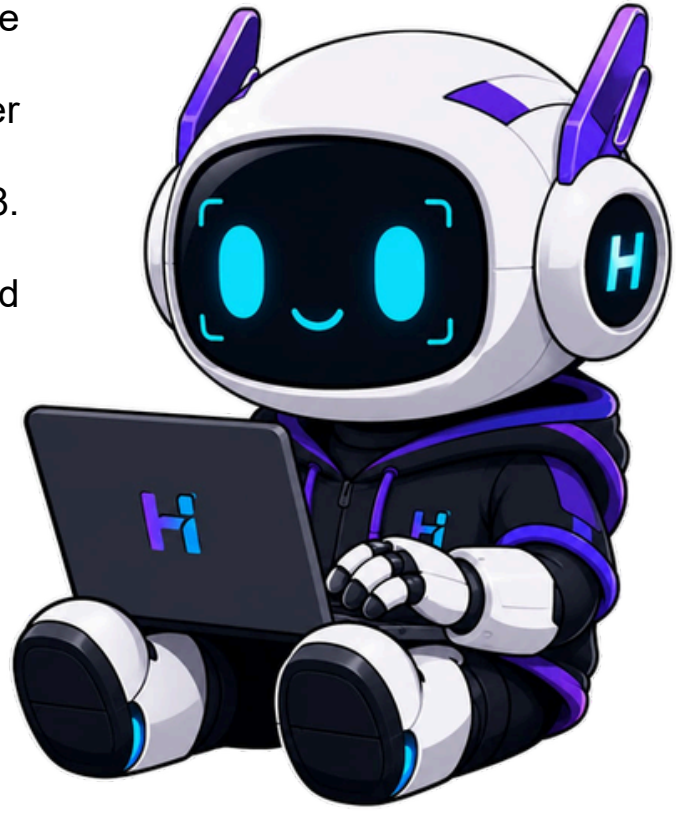
HYPERFUSION 2026

TITLE PAGE

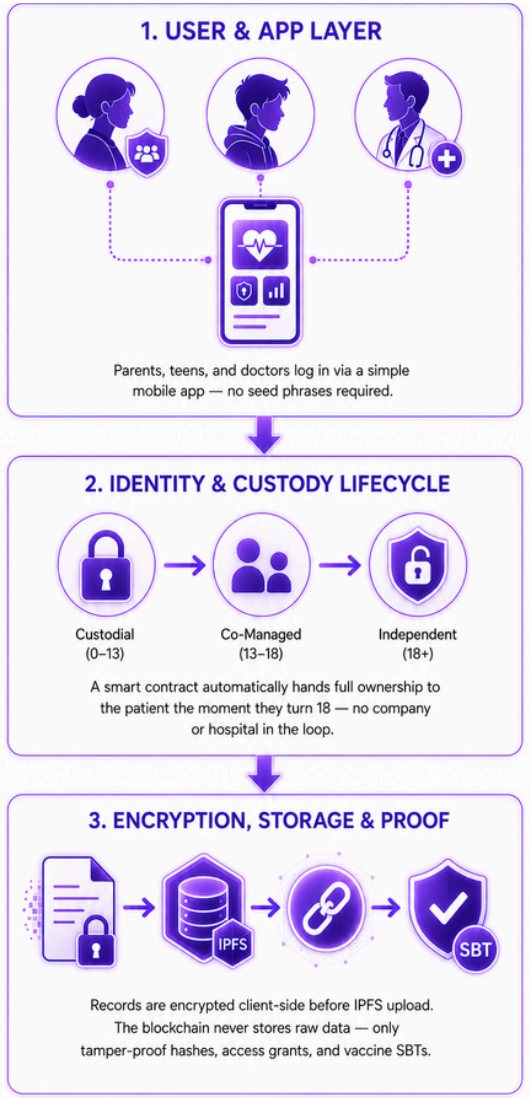
- **Problem Statement Title-** Modern healthcare is broken because patients don't own their medical data. Records are trapped in centralized hospital databases that are vulnerable to ransomware and fracture every time a patient switches providers. This forces reliance on easily lost paper documents and offers no secure way to transfer health record ownership from a parent to a child when they turn 18. As a result, individuals lack a portable, lifelong health identity and have zero control over their own life-saving medical information.
- **Theme-** Health-Tech
- **Team Name (Registered on Devfolio)-** dual-core



IDEA TITLE

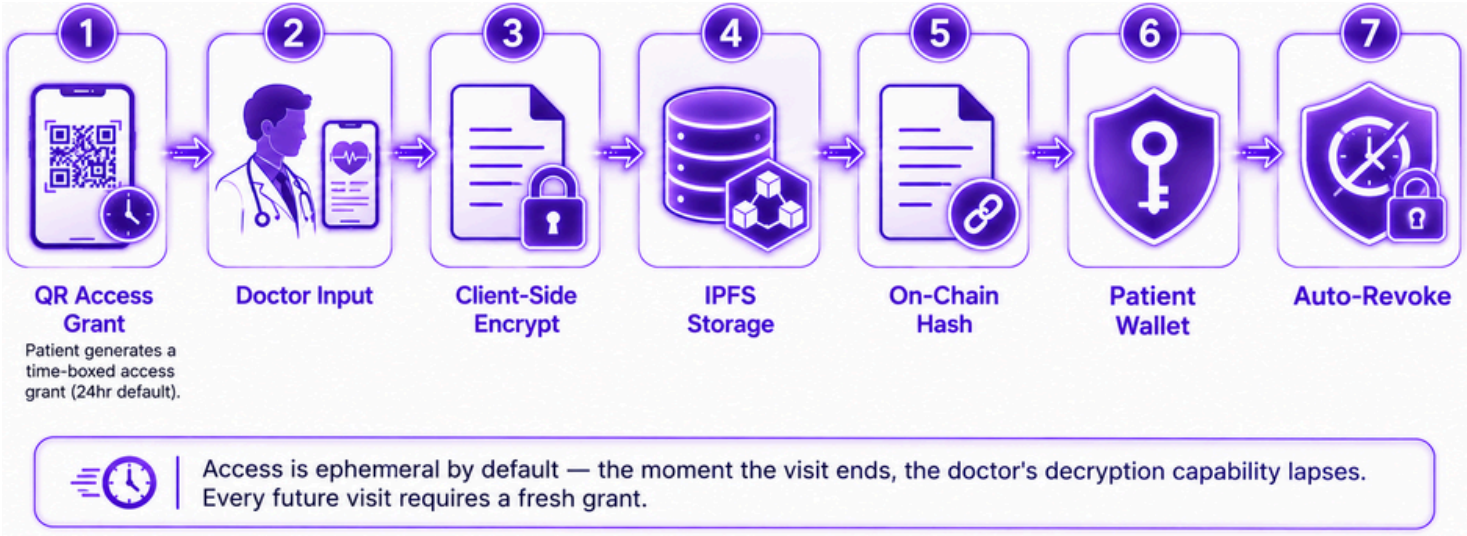
- **Proposed Solution (Describe your Idea/Solution/Prototype)**

- Detailed explanation of the proposed solution:
  - Patient-owned medical record system using Account Abstraction and IPFS.
  - Records are client-side encrypted; blockchain stores only tamper-proof hashes and access rules.
- How it addresses the problem:
  - Eliminates hospital honeypots, lost paper trails, and re-onboarding on provider switches.
  - Replaces vaccine cards with SBTs; grants doctors time-boxed, QR-based access.
- Innovation and uniqueness of the solution: "Progressive Sovereignty"—a smart contract autonomously transfers medical data ownership from parent to child at age 18, revoking guardian access automatically.



TECHNICAL APPROACH

- Technologies to be used:
  - Smart Contracts: Solidity, Foundry (for testing & fuzzing), ERC-1167 Minimal Proxies (gas-efficient patient identity clones).
  - Frontend: React.js (App Router), Tailwind CSS, wagmi / viem for blockchain interactions.
  - Identity & Auth: ERC-4337 Account Abstraction via Privy/Biconomy (gasless, email-based login—no seed phrases).
  - Storage & Security: IPFS via Web3.Storage, eth-crypto (ECIES client-side encryption before data leaves the device).
  - Blockchain Layer: Polygon PoS / Base L2 (sub-cent transaction costs and ~2-second finality).
- Methodology and process for implementation:



FEASIBILITY AND VIABILITY

Analysis of Feasibility

- Uses existing blockchain, IPFS and encryption technologies.
- Can be developed as a working prototype using low-cost L2 networks.
- Core loop—identity, records, visit, claim.
- Sensitive medical data remains off-chain, making the approach more practical and privacy-conscious.

Potential Challenges & Risks

- Private key loss without recovery.
- Blockchain transactions and smart-contract vulnerabilities.
- GDPR erasure vs. blockchain immutability.
- Interoperability with existing hospital EHR systems.
- Legal validity of automated custody transfer.

Strategies to Overcome Challenges

- Social recovery guardians plus optional Shamir's Secret Sharing key escrow.
- Foundry-based fuzz testing and third-party smart contract audits.
- Crypto-shredding: destroy per-record keys, leaving only orphaned on-chain hashes.
- Follow FHIR/HL7 standards for hospital EHR interoperability.
- Guardian attestation at creation plus legal review for jurisdictional recognition.

IMPACT AND BENEFITS

Potential Impact on Target Audience

- Gives new parents a secure custodial child record.
- Lets teens turning 18 inherit full ownership automatically.
- Gives relocating adults one portable, lifelong health identity.
- Improves emergency access to medical history instantly.

Benefits of the Solution

- Social: Better continuity of healthcare and patient empowerment.
- Economic: Reduces paperwork, duplicate tests, and administrative costs.
- Security: Minimizes dependence on centralized data repositories and improves access transparency.
- Environmental: Reduces paper-based medical records and documentation.



RESEARCH AND REFERENCES

- Details / Links of the reference and research work

https://drive.google.com/drive/folders/1xwBkBKiCuwsGY7dJMo3XHuySsuEJh51I?usp=sharing